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Indian Institute of Technology Hyderabad
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AI3403 | MULTI-AGENT SYSTEMS | PROJECT REPORT

NABH-NITI

Adaptive 3D Airspace Governance for High-Density UAV Traffic

A reproducible multi-agent simulation of locally negotiated traffic norms



NABH-NITI 3D airspace simulation dashboard.

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Abstract

An urban airspace used by delivery, inspection, cargo and emergency drones is a shared resource. When many autonomous aircraft request the same locations, a traffic controller must decide who moves, who waits and when a temporary restriction is justified. **NABH-NITI** studies this problem using a discrete-time, three-dimensional multi-agent simulation. The system represents unmanned aerial vehicles (UAVs) as agents with missions, deadlines, routes and priorities. Local sector agents monitor demand and, in the proposed governance mode, negotiate temporary traffic norms while an independent movement-safety layer enforces discrete collision-prevention rules.

The prototype compares three policies: fixed ID-based right-of-way, centralized utility-based allocation, and decentralized norm-adaptive allocation. Four negotiable norm types address medical urgency, congestion, directional flow and sector admission. A Python/FastAPI engine drives an interactive React/Three.js city that exposes UAV decisions, sector rules, event history and policy comparisons. The archived evaluation contains **1,080 distinct 28-tick trials** across three initial fleet sizes, six disruption scenarios and 20 random seeds. In those trials, NABH-NITI attained mean Jain fairness **0.9775**, compared with **0.9745** for centralized control; its mean 95th-percentile cumulative wait was **10.73** versus **11.14** ticks. The centralized baseline completed more missions on average (**157.55** versus **153.60**). No violations of the implemented discrete safety invariants were recorded. These results show measurable efficiency–fairness trade-offs in a research simulation; they are not evidence of real-flight safety or certified operational readiness.

Keywords: multi-agent systems, UAV traffic management, adaptive norms, decentralized governance, fairness, 3D A* search, discrete simulation.

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1 Introduction and Motivation

1.1 A shared-sky allocation problem

A city with autonomous medical, delivery, inspection, survey and cargo drones is not simply a collection of unrelated shortest-path problems. Two aircraft may have individually valid routes that request the same airspace cell at the same simulation step. A medical mission may be urgent, while several ordinary missions have already been waiting. Weather can close a region; high demand may cause queues; and local communication may be degraded. The difficult decision is how to allocate a scarce shared resource among competing autonomous agents.

For a concrete example, consider a delivery UAV and an emergency medical UAV attempting to enter the same virtual corridor. A fixed priority list gives a predictable answer but cannot react to the newly urgent mission. A global dispatcher can evaluate all candidates, but its decision process is centralized. NABH-NITI introduces a third possibility: nearby agents and a sector manager can propose an operational rule for a limited interval and then withdraw it when the situation changes.

Current discussions of unmanned aircraft traffic management (UTM) already recognize the importance of strategic coordination, route planning, dynamic restrictions and exchange of operational information [2, 1]. This project focuses on a narrower, controlled research question: *within a discrete simulator, how do locally negotiated norms compare with fixed and centralized policies on safety checks, service fairness, delay, mission completion and logical communication cost?*

1.2 Objectives and contributions

The project sets out to (i) represent a populated 3D airspace with heterogeneous UAV agents; (ii) implement the same separation checks under three allocation policies; (iii) introduce local norm formation, utility bids and fairness credits; (iv) support disruptive event injection; (v) expose internal agent decisions through a visual digital twin; and (vi) evaluate matched conditions using reproducible seeds.

Our main contribution is **adaptive sector-level traffic governance**: UAVs coordinate access to a shared airspace, and local agents introduce temporary rules when traffic conditions change. We combine this approach with a common movement-safety layer and an interactive 3D dashboard to compare how different policies behave.

2 Background and Research Position

2.1 Related concepts

NASA's UTM research discusses services including airspace organization, conflict management, weather restrictions and flight planning; the FAA describes a collaborative traffic-management ecosystem for low-altitude drone operations [1, 2]. India's Drone Rules and Digital Sky zoning provide important regulatory context for real drone operations, while the European U-space framework illustrates the need for clearly defined services and safety responsibilities [3, 4]. **Our four simulated norms are not Indian government regulations**, and the project does not interface with Digital Sky or issue real flight authorizations.

The routing component follows the classic A* family of shortest-path algorithms [5]. Fairness is quantified using Jain's index, originally developed for comparing allocations of a shared resource [6]. We apply these ideas to simulated UAV access opportunities rather than to continuous-flight control.

2.2 Where our experiment differs

Most background traffic-management concepts describe operational functions or service architectures; our experiment implements three deliberately simplified, directly comparable *allocation policies* inside the same simulation. Local sectors hold one temporary norm at a time, and affected UAVs give support votes. The accepted norm changes traffic scoring or admission but cannot disable the underlying safety shield.

3 Problem Formulation

3.1 Airspace graph and agents

The default backend grid has width $W = 24$, height $H = 20$, and $L = 4$ altitude layers, hence $|V| = WHL = \mathbf{1,920}$ addressable grid nodes. Its six-connected graph is $G = (V, E)$; allowed movements change one of the x, y, or z coordinates by one unit. Each edge has unit routing cost. A cell can be blocked by a temporary virtual weather restriction. The default sector assignment is

$$S(x, y, z) = \left(\left\lfloor \frac{x}{6} \right\rfloor, \left\lfloor \frac{y}{5} \right\rfloor\right), \tag{1}$$

which divides the standard map into 16 local sectors. Sector identifiers such as `S1-2` label these administrative regions; sectors are not separate geographic cities or independently deployed servers.

Each UAV i has a state comprising its ID, type, position $p_i(t)$, target g_i, a mission deadline, simulated battery reserve, waiting history, priority credits, planned path, and latest traffic bid. Five mission classes are implemented: *medical, delivery, inspection, survey, cargo*. Their modeled nominal priorities are 10, 2, 3, 2 and 1 respectively. A *tick* is one complete discrete update of traffic proposals, safe allocations, movement and counters; it does not correspond to a certified number of real seconds.

3.2 Allocation objective and shared constraints

At each tick, agents propose next-step cells. A policy selects a feasible set of movements. Performance is assessed through missions completed, cumulative tail waiting, service opportunity fairness and logical message counts, subject to the four principal simulated movement requirements shown in Table 1. The simulator does not maximize a single fixed weighted objective for all policies; instead, the policies use different ranking mechanisms whose *outcomes* are compared with the same metrics.

Table 1: Non-negotiable simulation safety invariants shared by all three policies.

Invariant	Meaning in the simulator
Exclusive occupancy	No two active UAVs may finish a tick in the same 3D node.
No edge swapping	Two UAVs may not exchange endpoints along the same edge in opposite directions in a single tick.
Restricted space	A UAV may not move into any grid node currently designated as closed.
Map boundaries	Positions must remain inside the configured 3D grid.

For two distinct UAVs i and j, the occupancy and head-on-swap rules can be written as

$$\begin{aligned} &p_i(t+1) \neq p_j(t+1), \\ &\quad \neg\big(p_i(t+1) = p_j(t) \ \wedge \ p_j(t+1) = p_i(t)\big), \qquad i \neq j. \end{aligned} \tag{2}$$

Here $p_i(t)$ is drone i's grid position at tick t. Its destination must also be an in-bounds, unblocked node. These constraints are checked after movement allocation, regardless of which governance policy ranked the requests.

4 Implemented System Architecture

The backend is written in Python. Its `core.py` implements the airspace, UAV states, routing, contention handling, candidate norm generation and safety checks. `experiments.py` runs repeatable matched-seed experiments and writes result tables. `api.py` exposes simulation commands and live state snapshots. The frontend uses React, TypeScript and Three.js through React Three Fiber; it receives events and state using WebSockets and displays a simulated city with clickable UAVs, active norms and performance measurements. Figure 1 shows the main separation of concerns.

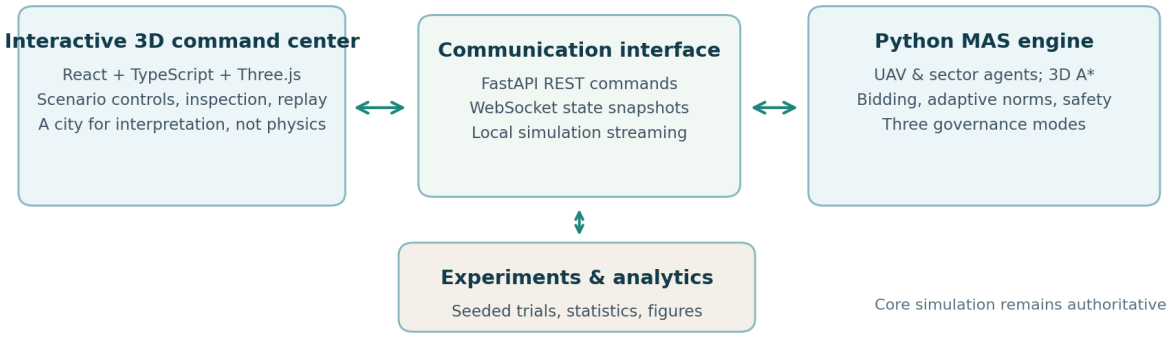


Figure 1: Separation between visualization, API transport, the authoritative simulation, and analysis. This architecture diagram is drawn from the implemented source modules.

The frontend's urban buildings, roads and fictional facilities make the abstract model observable during a demo. These decorations do not define the backend collision grid and must not be interpreted as an accurate geographical replica of Hyderabad or any other real city. The visualization includes five bundled GLB multirotor models and a smart display mode that shows a subset of the 50–500 simulated agents to avoid visual clutter. The reported *rendered* count may therefore be lower than the *simulated* count.

5 3D Routing and Safety Core

5.1 Shortest-path search

For a UAV at $s = (x_s, y_s, z_s)$ with target $g = (x_g, y_g, z_g)$, the algorithm uses A* with a Manhattan-distance heuristic:

$$f(n) = g(n) + h(n), \qquad h(n) = |x_n - x_g| + |y_n - y_g| + |z_n - z_g|. \tag{3}$$

Because the backend uses six-directional unit-cost moves, this is an admissible heuristic on the unobstructed grid. Blocked cells are excluded from legal neighboring nodes, and the search returns a graph path whenever a reachable path exists. This is *strategic topological routing*; it does not itself guarantee separation from other moving UAVs.

5.2 Time-step conflict resolution

After routing, the simulator receives proposed next nodes from all UAVs. When several drones request the same destination cell, the active policy ranks contenders. An independent movement-safety procedure then cancels any move into a cell whose current occupant will not vacate, and rejects head-on swaps. The cancellation pass repeats until the set of chosen movements is consistent. Rejected requests lead to holding/waiting updates; successful requests advance positions and adjust simulated battery and fairness state.

This layered implementation is a substantial engineering distinction: **finding a shortest geometric route is not the same as reserving a conflict-free movement at a particular tick.** The backend checks the resulting occupancy set after every step and raises a program assertion if its modeled hard invariants are broken.

5.3 Independent route test

The supplied `route_correctness_audit.json` records an independent comparison of A* against breadth-first search on **1,000 randomly obstructed, reachable grid cases**; all 1,000 cases passed the test's path-length and route-legality checks. This supports implementation correctness for the tested discrete shortest-path

cases only. It is not a 100% drone-prediction accuracy result, and it does not certify physical separation performance in real airspace.

6 Three Governance Approaches

We use the same airspace and safety rules under all three policies. Table 2 summarizes what is actually different.

Table 2: Implemented policy comparison, with benefits and costs.

Policy	How contested cells are allocated	Main trade-off
Fixed rules	Lower UAV numeric ID receives priority for a contested target. No temporary norms.	Very simple and zero modeled messaging; does not adapt this priority to urgency.
Centralized	One global ranking score reflects mission priority, deadline urgency and time waiting. No dynamic sector norms.	Coordinated ranking, but depends on global dispatch and the simulator counts a report and dispatch per agent per tick.
NABH-NITI	Local bids incorporate urgency, possible active norms and fairness credits; sector agents can activate temporary rules.	Adaptable priorities with local negotiation, but creates decision and communication overhead.

6.1 Utility scores are priorities, not monetary bids

For the centralized and baseline decentralized ranking, the Python code computes a base score

$$b_i^{(0)} = 3q_i + 7u_i(t) + 0.8\min(w_i, 15), \quad u_i(t) = \max\left(0, \frac{24 - (d_i - t)}{24}\right), \tag{4}$$

where q_i is mission priority, d_i is the deadline tick, and w_i counts consecutive waiting ticks. Fixed rules instead use $-\mathrm{ID}_i$ as the ranking bid. NABH-NITI may add $1.4c_i$ for fairness credits, a temporary rule bonus, and 8 for a selfish agent's inflated bid. In a blacked-out local sector the implemented bid is reduced to $0.75b_i^{(0)}$. These scores are deterministic *decision heuristics*; there is no real currency or deployed auction marketplace.

6.2 Fairness credits

Each UAV starts with five credits. Under NABH-NITI, when the fairness mechanism is enabled, the credit balance c_i follows

$$c_i(t+1) = \begin{cases} \min\{15,\ c_i(t) + 1\}, & \text{if drone } i \text{ holds,} \\ \max\{0,\ c_i(t) - 1\}, & \text{if drone } i \text{ moves.} \end{cases} \tag{5}$$

Credits give repeatedly delayed agents a larger bid bonus, helping prevent the same drone from losing access indefinitely. They affect priority ranking, not the hard movement constraints. A fairness-disabled ablation allows comparison with the full mechanism.

7 Adaptive Rule Engine and Sector Negotiation

7.1 Rule catalogue

Table 3: Four temporary operational norms and their exact model-level effect.

Rule	What the implemented code does	Possible downside
Medical priority	Adds a score bonus of 24 for medical UAVs in the affected sector.	Non-medical traffic can wait longer.
Congestion priority	Adds a waiting-dependent bonus up to 1.2×12 for sector traffic.	Improves ordering but cannot increase capacity.
One-way lane	On altitude index $z = 1$, prevents non-medical motion against the sector's chosen horizontal direction.	Some drones must reroute or hold; medical traffic is exempt.
Entry quota	Limits non-medical inter-sector admissions per tick, with a quota set from local agent count.	Traffic can queue outside the sector.

For an active entry-quota norm, the per-tick admission limit is

$$Q_s = \max\left(2, \left\lfloor \frac{N_s}{6} \right\rfloor\right), \tag{6}$$

where N_s is the number of UAVs in sector s when that norm is created. At most Q_s selected *non-medical* inter-sector entries are admitted per tick; medical UAVs are exempt.

7.2 Proposal, approval, expiry



Figure 2: Local normative-control lifecycle. Candidates are considered periodically and approved norms remain subordinate to the same safety layer as other policies.

At each fourth tick, a sector with no active rule and outside its cooldown window can propose a norm if local demand meets candidate conditions. The implementation considers medical traffic when a medical UAV is present and at least three agents share the sector. Congestion and directional proposals require larger local populations; entry quota requires at least 22 agents in the sector. Candidate scores depend on crowding, waiting, directional balance and a bounded descriptive rule-memory term. The sector collects programmed yes/no support votes from participating drones. For N_s local participants and votes $v_{i,s} \in \{0, 1\}$, the approval condition is

$$\alpha_s = \frac{1}{N_s} \sum_{i=1}^{N_s} v_{i,s}, \qquad \text{approve if } \alpha_s \geq 0.60. \tag{7}$$

A rejected proposal is recorded and followed by a brief cooldown. In the implementation, each vote depends on the UAV's mission type, waiting state, and selfish-behavior flag.

Accepted norms normally have **a 14-tick lifetime**, with one active norm per sector. After expiry or loss of sector communication, the rule is retired, and the sector waits three ticks before proposing again. For one-way lanes, an agent can align its chosen direction with an already active neighboring one-way norm. The log and rule history expose proposals, votes, activation, retirement and descriptive waiting observations. These before/after observations are *not causal proof* that a particular norm improved traffic.

7.3 Why keep hard constraints separate?

The agent collective can decide whether a lane should be one-way or medical traffic should receive a scoring advantage. It cannot decide to allow two aircraft into the same discrete node. This separation makes the research question coherent: adaptation changes access policy, while each policy operates within the same modeled safety envelope.

8 Stress Scenarios and Resilience

The prototype supports six stress-event families in the archived full experiment: congestion, medical, festival, storm, failure and selfish behavior. The UI also supports a normal operating state. Table 4 describes the actual simulation actions rather than suggesting physical flight capabilities that were not implemented.

Table 4: Actual synthetic disturbances included in the backend.

Scenario	Synthetic backend event
Congestion	Inject 110 additional mixed UAV agents during the run.
Medical	Introduce one urgent medical UAV; emergency missions use high nominal priority.
Festival	Inject 85 additional mixed UAV agents.
Storm	Close selected unoccupied grid cells within a central rectangular volume and clear stored routes; reopen later in the experiment.
Failure + blackout	Disable coordination in a chosen sector, remove one failed drone from the active set, and restore the link later.
Selfish agents	Mark approximately 40% of agents as strategic bidders, inflating their scores while safety checks remain unchanged.

A storm does *not* produce realistic wind loads, precipitation forces, or emergency landing. When the stored path is invalidated, a UAV attempts a new route; if no admissible next step can be reserved it waits. In the failure scenario the chosen UAV is removed from active traffic. A communications blackout is modeled as a degraded local decision channel, not a packet-level radio simulator.

9 Software Implementation and User Interface

9.1 Development stack

The backend uses Python, FastAPI and WebSockets. The interface uses React, TypeScript, Three.js, React Three Fiber and Vite. NumPy, Pandas and Matplotlib are used in experiment processing and figure creation; Pytest is used for automated backend checks. This is a local web application; it requires no physical UAVs, GNSS sensors or remote operations infrastructure.

9.2 Inspectability of decisions

The user can change governance policy (with a reset for comparable starting conditions), choose an initial population, enter a deterministic seed, inject events, inspect a UAV, follow its route, and open the active-norm and event panels. Research controls launch paired comparisons and fairness/norm ablations. The dashboard displays active fleet size, missions completed, tail waiting, fairness, messages, proposals and recorded discrete safety breaches.

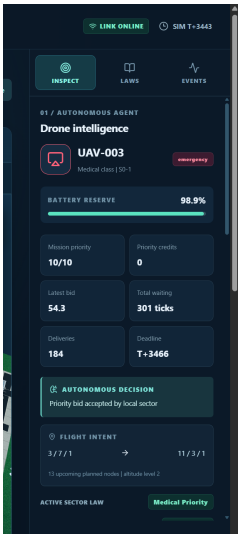
The visual renderer's smart mode purposely displays fewer models than the number of agents simulated. For example, a live display may say "65 rendered / 150 simulated." This is a visualization choice, not a reduction of the number of agents making decisions. The colored multirotor models are 3D assets; the project does not claim photorealistic or certified flight dynamics.



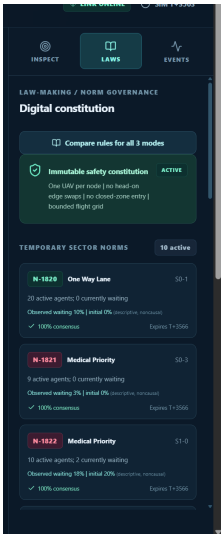
Figure 3: Actual project screenshot showing a weather-closure visualization during a running simulation. Pink wireframes identify visually restricted airspace; geometric detail comes from the digital twin.



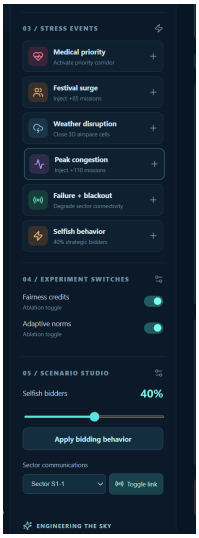
Figure 4: Actual local website: governance selection and fleet controls at left, procedurally modeled city and drones in the center, and live agent/law information at right. Screenshot used as an interface artifact, not as experimental evidence.



(a) UAV state, bid and flight intent.



(b) Fixed safety constitution and active norms.



(c) Synthetic disruption controls.

Figure 5: Three real UI captures illustrating agent transparency, governance and test controls. The narrow screenshots have been scaled to fit the page; the original images are part of the team archive.

10 Experimental Methodology

10.1 Reproducible experiment matrix

The archived file `results/full_1080_trials.csv` contains exactly 1,080 rows with no duplicated (policy, scenario, seed, initial population) combination. The factors are:

Factor	Settings recorded in the CSV
Governance	Fixed, centralized, NABH-NITI (3)
Starting fleet	50, 200, 500 UAVs (3)
Stress event	Congestion, medical, festival, storm, failure, selfish (6)
Random seed	Integers 0 through 19 (20)
Episode length	28 discrete simulation ticks
Total	$3 \times 3 \times 6 \times 20 = \mathbf{1,080}$

Thus each policy has 360 recorded trials. A fixed seed is used to initialize comparable randomized traffic, deadlines and target distributions for each matched scenario. It does not mean that all policies produce identical paths or outcomes. The reported means weight the recorded trial configurations equally; they are not population-weighted real-world projections. The saved batch does **not** contain a complete multi-seed ablation matrix: ablations are supported by a separate interactive experiment function but must not be represented as part of these 1,080 rows.

10.2 Evaluation metrics

- **Completed missions:** successful target arrivals/mission completions recorded over the episode.
- **Tail waiting:** 95th percentile of cumulative waiting ticks among active agents at the end of each run, then averaged over runs.
- **Service-opportunity fairness:** the Jain index of per-agent access ratios r_i among eligible non-medical

UAVs,

$$r_i = \frac{\text{moves}_i + 1}{\text{moves}_i + \text{waits}_i + 1}, \qquad J(r) = \frac{(\sum_{i=1}^{n} r_i)^2}{n \sum_{i=1}^{n} r_i^2}. \tag{8}$$

- **Logical messages:** abstract counts attributed by the policy logic to reports, dispatches, proposals and bids. They are not network packets or measured latency.
- **Potential conflicts:** attempted contention events or unsafe proposed movements resolved before execution; they are not actual in-air collisions.
- **Safety violations:** runtime failures of the discrete occupancy, swap and restriction invariants. A zero count does not establish continuous-space aviation safety.

11 Results and Discussion

11.1 Main cross-scenario comparison

Table 5 was calculated directly from the supplied archived trial CSV, not copied from a promotional dashboard. Values are per-episode means over 360 runs for each policy.

Table 5: Cross-scenario means from the archived 1,080 28-tick trials; 360 observations per policy. Smaller waiting and message counts are better, while higher completions/fairness are preferable.

Metric	Fixed	Centralized	NABH-NITI
Completed missions	**158.69**	157.55	153.60
Jain fairness	0.9734	0.9745	**0.9775**
95th-percentile wait (ticks)	11.19	11.14	**10.73**
Logical messages	**0**	15,365	4,476
Predicted potential conflicts	2,096.42	2,099.69	2,073.81
Safety-invariant violations	0	0	0

In this short-run dataset, NABH-NITI's mean fairness exceeds centralized control by about 0.0029 index points, and its mean p95 cumulative wait is about 0.41 ticks lower. It completes about 3.95 fewer missions per run on average (approximately 2.5% fewer than centralized). The policy therefore trades some mission completion throughput for slightly more equitable service and reduced tail waiting. Fixed rules has the highest mean completion count in this particular batch. **The data do not justify claiming NABH-NITI is best on every metric.**

The logical message counts are also informative, but only within the model: centralized mode is charged two messages per active UAV per tick, fixed mode none, and NABH-NITI is charged for local proposals and bids. As recorded, NABH-NITI counts roughly **71% fewer logical messages than centralized** across these runs. The difference is not a measured wireless bandwidth or deployment cost reduction; it reflects the explicit accounting scheme in the source code.

11.2 Effect of initial population

Figure 6 shows how the governance differences change with starting demand. At 500 initial UAVs, mean completed missions were 225.24 (fixed), 223.66 (centralized), and 212.85 (NABH-NITI), while the corresponding fairness values were 0.954, 0.958 and 0.965. The proposed policy's waiting advantage is clearer under this high-density configuration, but so is its completion cost. At 50 initial UAVs, completion counts are much closer between policies. These observations are more defensible than treating the 1,080 experiments as a single universal score.

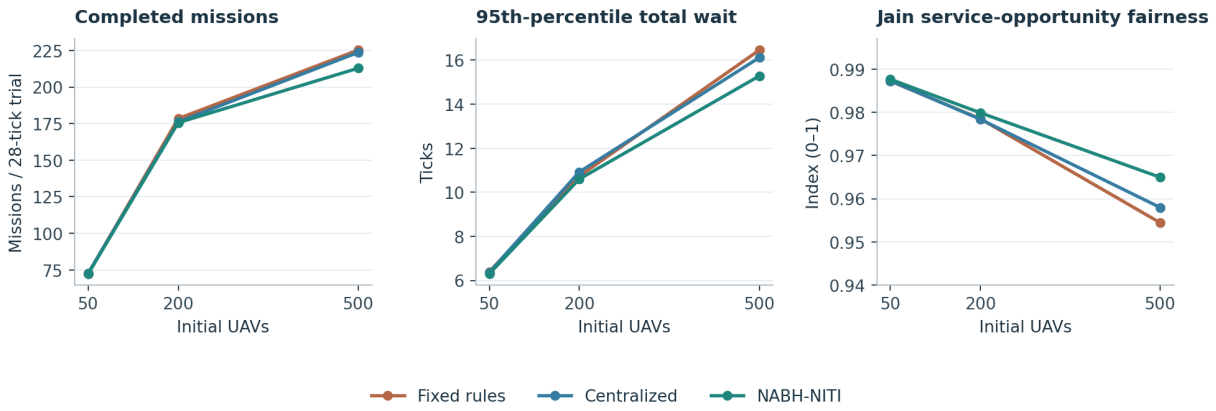


Figure 6: Mean outcomes across three initial fleet sizes from the archived full batch. All measurements refer to the same 28-tick episode length; chart points are descriptive averages.

11.3 Stress-scenario variation

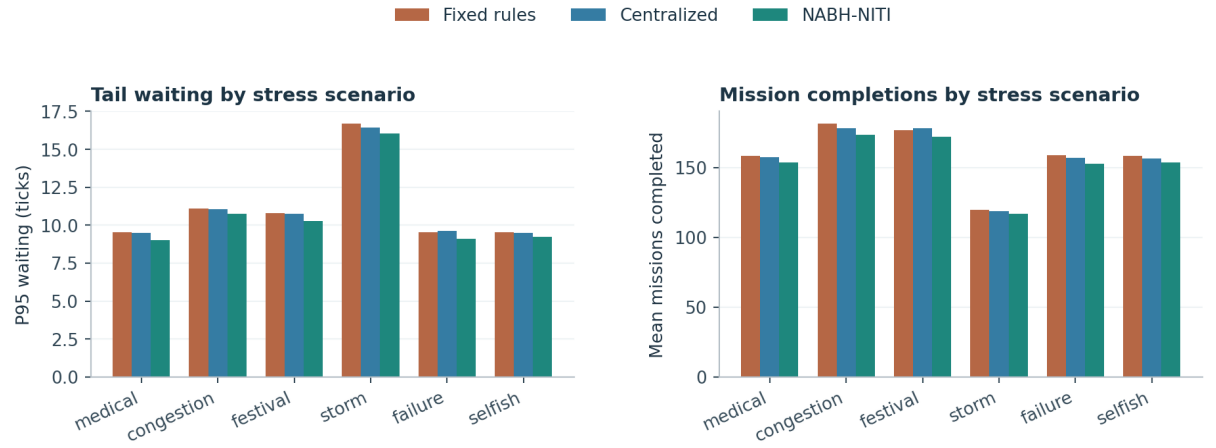


Figure 7: Per-scenario means for tail waiting and completed missions across the three policy modes. Scenarios are synthetic interventions applied using the same experiment schedule.

Storm runs produce the largest tail waiting values of these event groups (for example, mean p95 wait 16.02 ticks for NABH-NITI versus 16.42 for centralized). This is consistent with the simulation reducing available routing options; it is not a meteorological risk estimate. The medical, selfish, blackout and surge scenarios illuminate different trade-offs and support using scenario-specific analysis rather than one aggregate superiority claim.

11.4 Coordination overhead and ablations

The system can separately run NABH-NITI with fairness credits disabled and with adaptive norm creation disabled. These are useful ablation probes because they isolate the credit mechanism and the rule engine. The team's screenshots show one matched, interactive *90-tick, seed-7, 150-UAV* comparison including the two ablations. However, these single-run values are **not pooled with the 1,080-trial averages**; a comprehensive multi-seed ablation evaluation remains follow-up work.

These comparisons show how the three policies behave under the selected simulation conditions. Further experiments would be needed to assess longer-term behavior and determine whether the same differences appear in other settings.

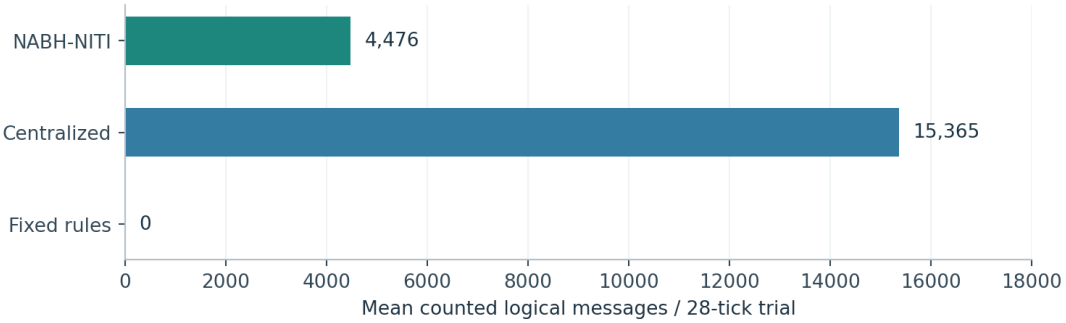


Figure 8: Mean *modeled logical message* counts from the same CSV; zero for Fixed Rules is a modeling convention, not evidence that deployed traffic management requires no communication.

12 Validation and Reproducibility

12.1 What was checked

The archived project includes Python automated tests for the simulation and API and a route audit against an independent breadth-first search reference. The source and project history report passing backend tests and interface-specific visualization checks; an up-to-date full browser performance study was not part of the experimental CSV. An operational website has also been demonstrated in team screenshots with live WebSocket telemetry. These form useful implementation evidence but should not be conflated with a measured aviation safety case.

For each trial, the backend records seed, scenario, policy, initial population, step count, completion metrics, waiting and fairness, proposals, rule counts, abstract message counts, and safety-invariant checks. It exports CSV and a summary JSON file, allowing readers to recompute all numeric figures in this report.

12.2 How to reproduce the experiment

1. Install Python requirements from `backend/requirements.txt`.
2. From the backend directory, execute the batch experiment command below.
3. Check the resulting CSV has 1,080 rows and examine the generated summary JSON.
4. For interactive replication, start the FastAPI server and the React frontend using the repository's run instructions.

```
python -m nabhniti.experiments --preset full --steps 28
      --workers 4 --out ../results/full_1080_trials.csv
```

Open-source implementation. The complete team code, reproducible experiment artifacts, tests and interactive 3D dashboard are available in the public repository below. The repository includes a clear setup guide and is referenced in this report for independent review.

Source-code repository:
https://github.com/koushik-varja/NABH-NITI

13 Limitations and Future Work

Several limitations remain. UAV motion is on an integer grid, not continuous three-dimensional dynamics; battery use is a simple proxy and not a model of aerodynamics, rotor thrust, payload mass or wind. Weather

is represented by restricted cells. Sector communication is a simplified availability flag and message counter, not a time-varying wireless link with packet loss, bandwidth or latency. Norm support is a programmed heuristic vote, not a formally verified strategic equilibrium. The city is procedural rather than GIS-validated or tied to India's real airspace permissions.

The completed batch is short: 28 ticks per trial. It offers informative comparisons under matched synthetic conditions but does not establish long-horizon stability, latency at high agent counts, general policy optimality, or a trained learning-based controller. The zero safety-invariant count describes a particular discrete reservation model and does not translate into a real-world risk figure. Browser frame rates at 500 UAVs were not measured as part of the archived research dataset.

Concrete next steps are longer paired-seed studies and confidence intervals, multi-seed ablations, more realistic comms and energy models, explicit landing/contingency procedures, continuous trajectory separation, official airspace data integration, and evaluation of camera/render performance under representative hardware. Any physical use would require flight-testing, real risk assessment, regulatory permissions and a much more detailed safety argument.

14 Conclusion

NABH-NITI demonstrates a complete, inspectable experimental setup for studying collective traffic governance in a shared, synthetic urban airspace. The implemented agents route through a 3D grid, compete for discrete movement slots, and, in the proposed approach, negotiate temporary sector norms that address emergency access, congestion and directional traffic. The independent safety checks remain the same for all three governance policies.

The 1,080-run archive shows small improvements in fairness and tail waiting for the proposed policy compared with centralized control, alongside fewer completed missions. The experimental value lies in making this trade-off visible, reproducible and explainable. The project supports the central Multi-Agent Systems lesson: **individual agent goals, local coordination, fairness and non-negotiable shared constraints must be designed together**, not treated as separate animation features.

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Individual Contributions

Member and roll number	Technical responsibility and specific contribution
Varja Krishna Koushik AI23BTECH11031	**3D routing and safety core.** Airspace graph, six-neighbor A* pathfinding, UAV movement and mission logic, contention resolution, blocked-cell routing, occupancy/head-on-swap checks and the Fixed Rules baseline. These modules provide the shared, safety-constrained movement foundation used in every policy comparison.
Shreya Ratnagiri ES24BTECH11031	**Adaptive sector-norm engine.** Sector traffic observation and norm candidacy, proposal/vote evaluation, four operational rule types, activation and 14-tick expiry, sector cooldowns, rule history and its interaction with non-negotiable movement constraints. This is the subsystem through which the proposed governance adapts over time.
Mangamuri Venkata Pranati CS23BTECH11034	**Negotiation and fairness.** Utility-based bids, competing movement allocation, deadline/wait priority, fairness-credit adjustments, the centralized ranking benchmark and fairness/norm ablation comparisons.
Rashmika Mandapalli CS23BTECH11033	**Scenario and resilience testing.** Medical, festival, congestion, weather, UAV-failure and blackout events; selfish bidding scenarios; disruption logic and stress-case testing.
Bala Antony Poojitha CS23BTECH11020	**Experiments and integration.** Seeded batch execution, the 1,080-run results archive, evaluation metrics, policy comparisons, result plots, backend-to-UI data integration and verification workflows.
Dhanavath Mohan Babu CS23BTECH11014	**3D command center.** React/Three.js city and aircraft visualization, drone inspection and route highlights, mission dashboard, camera modes, presentation views and replay interaction.